what's new in ruby 4.0 ?

Now I have comprehensive information about Ruby 4.0. Let me create a detailed report for this technical user.

Ruby 4.0: Major Features and Technical Advancements

Ruby 4.0 launched on Christmas Day 2025, marking the language's 30th anniversary with substantial architectural improvements and performance enhancements.[1] [2] As an infra-oriented developer, you'll likely find the isolation and JIT capabilities particularly relevant to production environments and infrastructure automation.

Architectural Innovations

ZJIT: A New Generation Just-In-Time Compiler

ZJIT represents a fundamental shift in Ruby's compilation strategy. Unlike YJIT's lazy basic block versioning approach, ZJIT employs a more traditional method-based compilation architecture with SSA (Static Single Assignment) IR and larger compilation units.[1] [2] This design philosophy prioritizes contributor accessibility—the "textbook" compiler design is intentionally more modular to encourage community contributions from optimization specialists.

Performance-wise, ZJIT currently exceeds the interpreter but lags behind YJIT in production scenarios. The team explicitly recommends deferring production deployment until Ruby 4.1, when ZJIT is expected to surpass YJIT's performance.[3] [2] Rust 1.85.0+ is required for ZJIT support; enable it with the `--zjit` flag at runtime or via `RubyVM::ZJIT.enable`.

Ruby::Box: In-Process Isolation

Ruby::Box introduces an experimental isolation mechanism for definitions within a single process—a game-changing feature for infrastructure use cases. You can load code in separate "boxes" with isolated global variables, constants, class definitions, and loaded native libraries.[1] [2]

Key expected use cases include:

- Test isolation without affecting global state (critical for CI/CD pipelines)
- Blue-green deployment within a single application server instance
- Dependency upgrade evaluation with response comparison
- Foundation layer for future "package" abstractions[2]

Enable with `RUBY_BOX=1` environment variable. Given the language around "foundation (low-level) API," expect higher-level abstractions in future releases. This is particularly valuable for complex

infrastructure scenarios where multiple versions of the same library need coexistence.

Language and Performance Improvements

Syntax Refinement: Leading Logical Operators

Logical operators (`||`, `&&`, `and`, `or`) now continue from the previous line when positioned at line start, mirroring the fluent dot operator behavior.[1] [2] [4] This addresses a long-standing formatting preference and reduces diff noise when managing complex conditionals—particularly relevant for infrastructure-as-code maintainability.

Splat Behavior Consistency

`*nil` no longer calls `nil.to_a` (aligning with `**nil` not calling `nil.to_hash`).[1] [2] This subtle change eliminates unnecessary array allocations in splat operations, improving efficiency in array unpacking operations.

Implementation Performance Optimizations

The core improvements deliver measurable gains:

- `Class#new` is significantly faster, especially with keyword arguments (integrated into both YJIT and ZJIT)
- Garbage collection heap pools now grow independently, reducing memory usage for variable-lifetime object distributions[2]
- Instance variable access uses a new internal "fields" object for generic ivar objects (String, Array, TypedData)
- `object_id` and `hash` are faster on Class/Module objects
- Symbol table and frozen strings use lock-free hash sets[2]

Ractor (Parallel Execution) Refinements

Ractor received substantial stability and performance improvements targeting experimental-to-stable graduation:

- **New `Ractor::Port` class** replaces the previous `Ractor.yield`/`Ractor#take` API with clearer IPC semantics (`receive`, `send`, `close`, `closed?`)
- **Reduced global lock contention**: Method cache lookups avoid locking in most cases; class instance variable access is faster and lock-free
- **Improved CPU cache behavior**: Per-ractor object allocation counters and thread-local memory allocation counters reduce cache contention[5] [2]
- `Ractor#join` **and** `Ractor#value` provide Thread-like behavior for Ractor termination waiting
- `Ractor.shareable_proc` **and** `Ractor.shareable_lambda` simplify sharing logic between parallel execution units[2]

These changes represent substantial engineering effort toward removing Ractor's experimental designation in Ruby 4.1.

Core Library Evolution

Standard Library Promotions

`Set` and `Pathname` migrated from default gems to core classes, improving integration and eliminating external dependency management. `Set` now uses simpler `inspect` output (`Set[1, 2, 3]` vs. `#<Set: {1, 2, 3}>`).[5] [2] Note: `SortedSet` has been removed; use the external `sorted_set` gem if needed.

Debugging and Introspection Enhancements

- `instance_variables_to_inspect`: Control which instance variables appear in `inspect` output, reducing noise from memoization variables[1]
- Extended `source_location` returns 5 elements instead of 2, enabling precise IDE/tooling integrations
- Enhanced `ErrorHighlight`: ArgumentError now displays both caller and callee code snippets with exact locations[3] [2]

String and Math Refinements

- `String#strip` and variants accept selector arguments for precise character removal (including character ranges)
- `Math.log1p` and `Math.expm1` added for numerically stable small-number calculations
- Unicode upgraded to version 17.0, including Emoji 17.0

Socket and I/O Improvements

- `Socket.tcp` and `TCPSocket.new` accept `open_timeout` keyword argument
- `File::Stat#birthtime` now available on Linux via statx syscall
- `IO.select` accepts `Float::INFINITY` as timeout[3]

Breaking Changes and Removals

Several long-deprecated features were removed:

- `Ractor.yield`, `Ractor#take`, `Ractor#close_incoming`, `Ractor#close_outgoing` (replaced by `Ractor::Port`)
- `Kernel#open` and `IO` process creation with leading `|` (removed after Ruby 3.3 deprecation)
- `Process::Status#&` and `#>>` operators
- RJIT moved to external library; only YJIT and ZJIT remain[2]

The CGI library is no longer provided in full—only `cgi/escape` and related methods persist.[2]

Ecosystem Updates

RubyGems and Bundler upgraded to version 4, with numerous default gems refreshed. Prism 1.7.0 adoption continues improving Ruby parsing infrastructure. OpenSSL updated to 4.0.0, and error_highlight enhanced for debugging support.[2]

Strategic Implications

Ruby 4.0 positions the language for infrastructure-scale parallelism via Ractor maturation and isolation via Box. The ZJIT foundation invites community compiler optimization—valuable if you're running custom Ruby implementations or considering contributions. The performance baseline improvements (GC, object allocation, variable access) benefit all applications without configuration changes.

For infrastructure-as-code and automation workflows, the logical operator syntax improvement and Ruby::Box isolation merit attention. Production ZJIT adoption should wait for Ruby 4.1; YJIT remains the recommended JIT for current deployments.[3] [2]

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1. https://blog.saeloun.com/2025/12/24/what-is-new-in-ruby-4/
2. https://www.ruby-lang.org/en/news/2025/12/25/ruby-4-0-0-released/
3. https://tech.yahoo.com/computing/articles/ruby-4-0-brings-zjit-202222203.html
4. https://rubyreferences.github.io/rubychanges/4.0.html
5. https://linuxiac.com/ruby-4-0-released-with-ruby-box-isolation-and-new-zjit-compiler/
6. https://www.jstor.org/stable/585686?origin=crossref
7. https://arxiv.org/pdf/2204.05345.pdf
8. http://arxiv.org/pdf/2503.19238.pdf
9. https://fedoraproject.org/wiki/Changes/Ruby_4.0
10. https://blog.codeminer42.com/whats-new-in-ruby-4-0/
11. https://www.reddit.com/r/ruby/comments/1pvzrsd/ruby_changes_ruby_40_annotated_changelog/
12. https://devcenter.heroku.com/changelog-items/3521
13. https://lwn.net/Articles/1051953/
14. https://www.semanticscholar.org/paper/173c1d9d07b06b30f24fac0a44db2bf40031f4cf
15. https://peerj.com/preprints/2758v1
16. https://linkinghub.elsevier.com/retrieve/pii/S000398619790182X
17. https://ashpublications.org/blood/article/106/11/2572/120880/A-Phase-I-Humanized-AntiCD40-Monoclonal-Antibody